

Name _____ Period _____

Skill 18.1 Exercise 1

Refer to the dictionary below,

```
zodiac_elements = {"water": ["Cancer", "Scorpio", "Pisces"], "fire": ["Aries",  
"Leo", "Sagittarius"], "earth": ["Taurus", "Virgo", "Capricorn"], "air":["Gemini",  
"Libra", "Aquarius"]}
```

Print out the list of zodiac signs associated with the "earth" element.

```
print(zodiac_elements["earth"])
```

Print out the list of the "fire" signs.

```
print(zodiac_elements["fire"])
```

Skill 18.2 Exercise 1

Review the code below and indicate what is printed.

```
zodiac_elements = {"water": ["Cancer", "Scorpio", "Pisces"], "fire": ["Aries",  
"Leo", "Sagittarius"], "earth": ["Taurus", "Virgo", "Capricorn"], "air":["Gemini",  
"Libra", "Aquarius"]}  
print(zodiac_elements["energy"])
```

```
key error; because "energy" is not a key in zodiac_elements.
```

Add an if statement to fix the error.

```
if "energy" in zodiac_elements:  
    Print(zodiac_elements["energy"])
```

Add the key "energy" to the zodiac elements. It should map to a value of "Not a Zodiac element".

```
zodiac_elements["energy"] = "Not a Zodiac element"
```

Name _____ Period _____

Skill 18.3 Exercise 1

Imagine you're analyzing survey responses stored as dictionaries. Some responses are missing values, which is very common in real data.

```
survey_responses = [  
    {"age": 25, "income": 50000, "city": "Seattle"},  
    {"age": 32, "income": 62000},           # missing city  
    {"age": 45, "city": "Chicago"},        # missing income  
    {"age": 29, "income": 54000, "city": "Austin"}  
]
```

Iterate over the *survey_responses* and print out the value of *city* for each response. If there is no city store, print *Unknown*.

```
for s in survey_responses:  
    city = s.get("city", "Unknown")  
    print(city)
```

Iterate over the *survey_responses* and “safely” calculate the average income.

```
count = 0  
total_income = 0  
for s in survey_responses:  
    if(s.get('income') != None):  
        total_income += s.get('income')  
        count += 1  
  
avg_income = total_income/count  
print(avg_income)
```

Name _____ Period _____

Skill 18.4 Exercise 1

Refer to the dictionary below,

```
student = {  
    "name": "Alex",  
    "age": 20,  
    "major": "Data Science",  
    "gpa": 3.7  
}
```

Indicate the output for each call below,

Call	Output
<code>print(student.get("name"))</code>	Alex
<code>print(student.get("year"))</code>	None
<code>print(student.get("year", "Not provided"))</code>	Not provided
<code>print(student.get("gpa", 0.0))</code>	3.7
<code>value = student.pop("age") print(value)</code>	20
<code>print(student)</code>	{name: Alex, major: Data Science, gpa: 3.7}
<code>print(student.pop("minor", "None declared"))</code>	None declared
<code>print(student)</code>	{name: Alex, major: Data Science, gpa: 3.7}
<code>print(student.pop("gpa"))</code>	3.7
<code>print(student)</code>	{name: Alex, major: Data Science}
<code>print(student.pop("age"))</code>	None

Name _____ Period _____

Skill 18.4 Exercise 2

You are designing the video game Big Rock Adventure. Below is a dictionary of items that are in the player's inventory which add points to their health meter.

```
available_items = {"health potion": 10, "cake of the cure": 5, "green elixir": 20,  
"strength sandwich": 25, "stamina grains": 15, "power stew": 30}  
health_points = 20
```

In one line, add the corresponding value of the key "stamina grains" to the *health_points* variable and remove the item "stamina grains" from the dictionary. If the key does not exist, add 0 to *health_points*.

```
health_points += available_items.pop("stamina grains", 0)
```

In one line, add the value of "mystic bread" to *health_points* and remove the item from the dictionary. If the key does not exist, add 0 to *health_points*.

```
health_points += available_items.pop("mystic bread", 0)
```

What is value of *health_points* after the 2 lines of code above run? What are the contents of *available_items*?

15

```
available_items = {"health potion": 10, "cake of the cure": 5, "green elixir": 20,  
"strength sandwich": 25, "power stew": 30}
```

Skill 18.5 Exercise 1

Imagine you pulled one row of data from an API and stored it as a dictionary. Before analyzing it, you want to see what variables (columns) are available.

```
survey_response = {  
    "age": 34,  
    "income": 72000,  
    "education": "Bachelor's",  
    "city": "Denver"  
}
```

Use the *keys* method to explore the column names.

```
column_names = survey_response.keys()
```

```
for c in column_names:  
    print(c)
```

Name _____ Period _____

Below is a list of `expected_fields`,

```
expected_fields = ["age", "income", "education", "city", "state"]
```

Write code that:

- Creates an empty list called *missing_fields*
- Loops through *expected_fields*
- Uses the dictionary's *keys* method to check if each field is present
- Adds any missing field names to *missing_fields*
- Prints the final list of missing fields

```
missing_fields = []  
for f in expected_fields:  
    if(f not in survey_response.keys()):  
        missing_fields.append(f)  
  
print(missing_fields)
```

Skill 18.6 Exercise 1

Imagine you collected survey data about the number of hours people exercise per week:

```
exercise_data = {  
    "Alice": 5,  
    "Bob": 3,  
    "Charlie": 7,  
    "Diana": 4  
}
```

Use the dictionary's `values()` method to access all exercise hours.
Compute the average number of hours people exercise per week.
Print the result.

```
total = 0  
  
for value in exercise_data.values():  
    total += value  
avg = total/len(exercise_data)  
print(avg)
```

Name _____ Period _____

Skill 18.7 Exercise 1

Refer to the dictionary below,

```
pct_women = {"CEO": 28, "Engineering Manager": 9, "Pharmacist": 58, "Physician": 40, "Lawyer": 37, "Aerospace Engineer": 9}
```

Use a for loop to iterate through the items of *pct_women*. For each key : value pair, print out a string that looks like,

Women make up {value} percent of {key}s.

```
pct_women = {"CEO": 28, "Engineering Manager": 9, "Pharmacist": 58, "Physician": 40, "Lawyer": 37, "Aerospace Engineer": 9}

for k,v in pct_women.items():
    result = "Women make up {percent} percent of {profession}s".format(percent = v, profession = k)
    print(result)
```